

Embedding Model Comparison — Top-k Retrieval

all-MiniLM-L6-v2 vs. multilingual-e5-small · 50 passages, 40 queries across 10 categories · English-only, mixed general knowledge · generated 2026-07-21T07:03:01 UTC

1. Executive summary & recommendation

- **Recommended: all-MiniLM-L6-v2** — wins 5 of 8 head-to-head metrics on this English-only corpus.
- Both models achieve near-ceiling ranking quality here, so the deciding factors are **score separability** (how safely you can set a similarity cutoff) and operational cost.
- Because the two models live on different cosine scales, each must use its **own** calibrated threshold τ (see calibration section) — never share one cutoff between them.

Metric	all-MiniLM-L6-v2	multilingual-e5-small	Winner
nDCG@10	0.979	0.975	all-MiniLM-L6-v2
Recall@5	0.980	0.990	multilingual-e5-small
MRR	0.985	0.971	all-MiniLM-L6-v2
Score separability	0.300	0.059	all-MiniLM-L6-v2
Average Precision	0.847	0.817	all-MiniLM-L6-v2
Dynamic-k F1	0.358	0.366	multilingual-e5-small
False-answer rate	0.167	0.000	multilingual-e5-small
Query latency (ms)	1.541	4.953	all-MiniLM-L6-v2

2. Setup

Each model uses its own prefix convention and L2-normalized cosine similarity. Same corpus, queries, qrels, and k grid for both. Run on CPU.

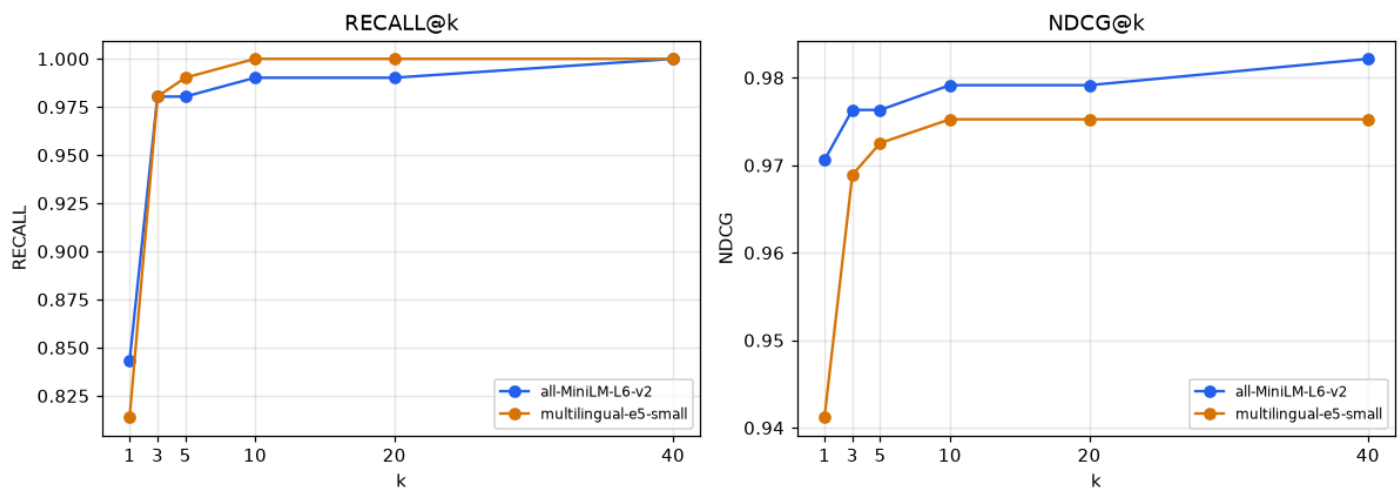
	all-MiniLM-L6-v2	multilingual-e5-small
Params	~22M	~118M
Embedding dim	384	384
Query prefix	(none)	'query: '
Passage prefix	(none)	'passage: '
Corpus encode / item (ms)	3.86	17.67
Query encode / item (ms)	1.54	4.95

Fairness note. `multilingual-e5-small` requires `query: / passage: prefixes`; `all-MiniLM-L6-v2` uses none. Both are applied automatically. The corpus is **English-only**, so e5's multilingual strength is *not* exercised here — this is a pure English semantic-retrieval comparison.

3. Aggregate metrics

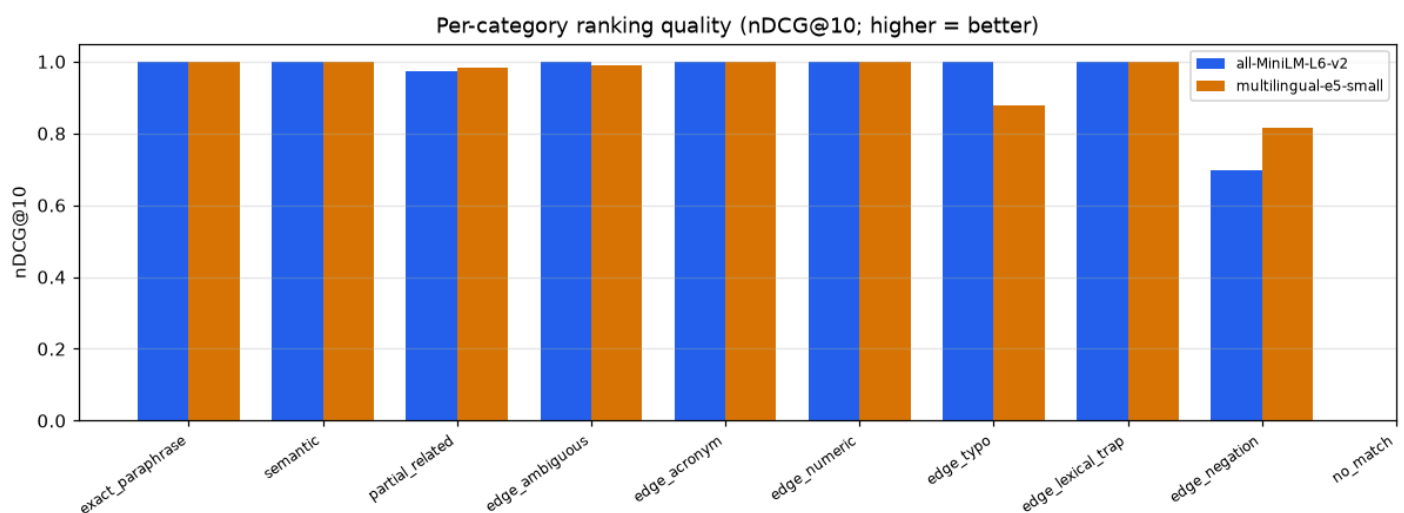
Averaged over answerable queries; best value per row highlighted. k grid: [1, 3, 5, 10, 20, 40].

Metric	all-MiniLM-L6-v2	multilingual-e5-small
RECALL@1	0.843	0.814
RECALL@3	0.980	0.980
RECALL@5	0.980	0.990
RECALL@10	0.990	1.000
RECALL@20	0.990	1.000
RECALL@40	1.000	1.000
PRECISION@1	0.825	0.800
PRECISION@3	0.367	0.367
PRECISION@5	0.220	0.225
PRECISION@10	0.113	0.115
PRECISION@20	0.056	0.058
PRECISION@40	0.029	0.029
HIT@1	0.971	0.941
HIT@3	1.000	1.000
HIT@5	1.000	1.000
HIT@10	1.000	1.000
HIT@20	1.000	1.000
HIT@40	1.000	1.000
NDCG@1	0.971	0.941
NDCG@3	0.976	0.969
NDCG@5	0.976	0.973
NDCG@10	0.979	0.975
NDCG@20	0.979	0.975
NDCG@40	0.982	0.975
MRR	0.985	0.971
Separability	0.300	0.059



4. Per-category performance

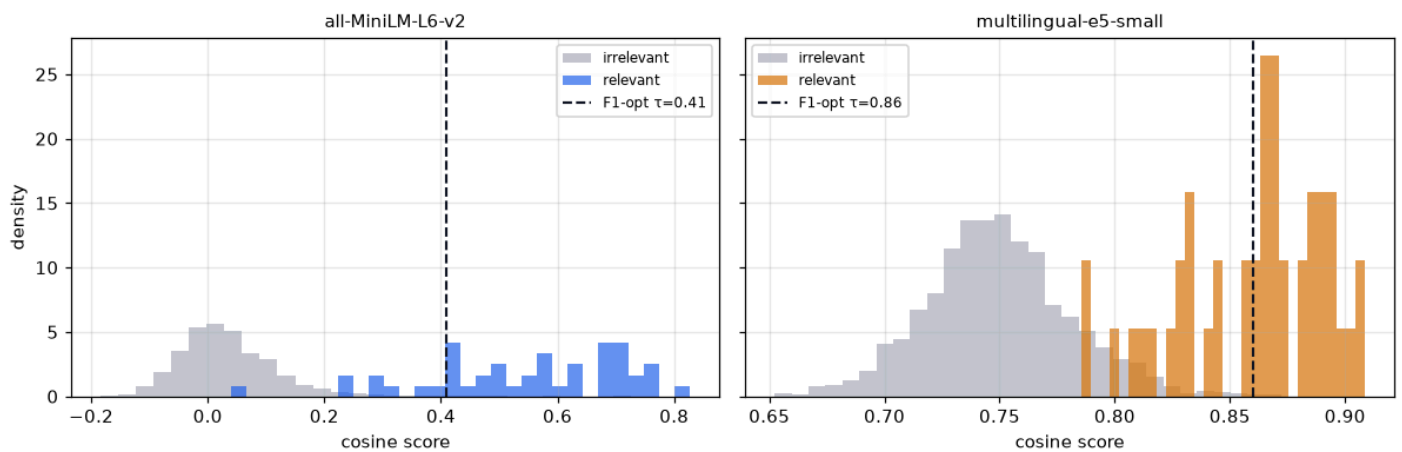
Where each model wins or loses, especially on the adversarial edge cases.



5. Score-distribution & separability

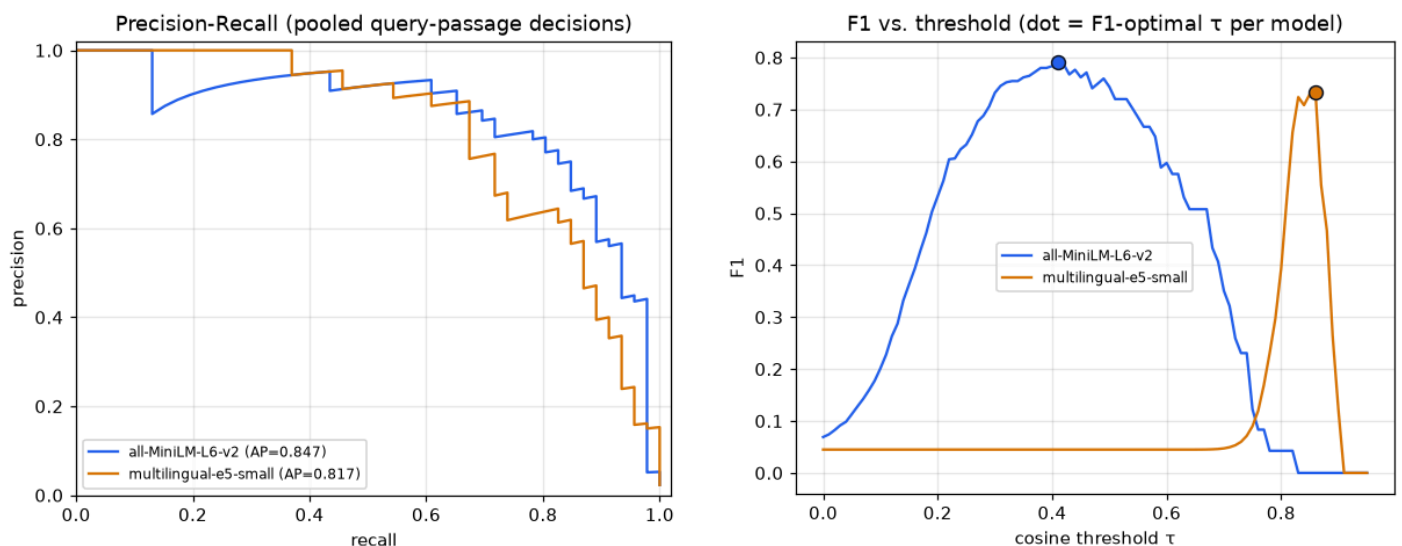
How well each model separates relevant from irrelevant chunks. A wide gap means a similarity cutoff is safe; overlapping distributions mean thresholding is risky. The dashed line is each model's F1-optimal τ .

Score distributions: relevant vs. irrelevant (separation = threshold safety)



6. Threshold calibration & dynamic-k policy

Per-model cosine cutoffs (they differ because the models sit on different score scales) and the deployable retrieval rule: keep chunks with score $\geq \tau$, clamped to [5, 40] chunks.



Operating point	all-MiniLM-L6-v2	multilingual-e5-small
f1_optimal	$\tau=0.41 \cdot P=0.760 \cdot R=0.826 \cdot F1=0.792$	$\tau=0.86 \cdot P=0.879 \cdot R=0.630 \cdot F1=0.734$
high_precision@0.9	$\tau=0.49 \cdot P=0.909 \cdot R=0.652 \cdot F1=0.759$	$\tau=0.87 \cdot P=0.947 \cdot R=0.391 \cdot F1=0.554$
high_recall@0.8	$\tau=0.42 \cdot P=0.771 \cdot R=0.804 \cdot F1=0.787$	$\tau=0.83 \cdot P=0.644 \cdot R=0.826 \cdot F1=0.724$
Dynamic-k policy [5–40]	$\tau=0.41 \cdot P=0.220 \cdot R=0.957 \cdot F1=0.358$ mean chunks=5.00 (range 5–5) · false-answer rate=0.167	$\tau=0.86 \cdot P=0.225 \cdot R=0.978 \cdot F1=0.366$ mean chunks=5.00 (range 5–5) · false-answer rate=0.000

The **dynamic-k precision** is bounded by the $k_{\min}=5$ floor: when a query has only 1–2 relevant passages, returning at least 5 chunks necessarily includes irrelevant ones. The floor is a deliberate trade to avoid starving the LLM context; recall and the false-answer rate are the more meaningful numbers for this policy.

7. Per-query side-by-side deep dive

Top-5 retrieved chunks per model with cosine scores. Green = a relevant (gold) passage. Ordered by category; queries where the models disagree on the #1 result are badged.

exact_paraphrase **How do plants make their own food using sunlight?**

all-MiniLM-L6-v2 gold rank: 1 · multilingual-e5-small gold rank: 1

#	all-MiniLM-L6-v2	multilingual-e5-small
1	p01 0.672 Photosynthesis is the process by which green plants and some other org	p01 0.886 Photosynthesis is the process by which green plants and some other org
2	p380.395 Solar panels convert sunlight directly into electricity using photovol	p380.872 Solar panels convert sunlight directly into electricity using photovol
3	p400.288 Honeybees are insects that pollinate flowering plants as they collect	p340.835 Tea is an aromatic beverage made by pouring hot water over cured leave
4	p020.183 The water cycle describes how water evaporates from oceans and lakes,	p020.828 The water cycle describes how water evaporates from oceans and lakes,
5	p390.172 Recycling is the process of collecting and reprocessing used materials	p400.821 Honeybees are insects that pollinate flowering plants as they collect

exact_paraphrase

When did humans first walk on the Moon?

all-MiniLM-L6-v2 gold rank: 1 · multilingual-e5-small gold rank: 1

#	all-MiniLM-L6-v2	multilingual-e5-small
1	p09 0.741 On 20 July 1969, the Apollo 11 mission landed the first humans on the	p09 0.889 On 20 July 1969, the Apollo 11 mission landed the first humans on the
2	p50 0.260 In 1903 the Wright brothers achieved the first controlled, powered, su	p50 0.781 In 1903 the Wright brothers achieved the first controlled, powered, su
3	p12 0.214 NASA, the National Aeronautics and Space Administration, is the United	p10 0.777 Mars is the fourth planet from the Sun and is often called the Red Pla
4	p10 0.193 Mars is the fourth planet from the Sun and is often called the Red Pla	p06 0.773 According to Albert Einstein's theory of relativity, the speed of ligh
5	p05 0.130 Gravity is the force of attraction between masses. Isaac Newton descri	p36 0.773 The modern Olympic Games are an international sporting event held ever

exact_paraphrase

What is the tallest mountain on Earth?

all-MiniLM-L6-v2 gold rank: 1 · multilingual-e5-small gold rank: 1

#	all-MiniLM-L6-v2	multilingual-e5-small
1	p19 0.703 Mount Everest, on the border of Nepal and Tibet, is the highest mounta	p19 0.880 Mount Everest, on the border of Nepal and Tibet, is the highest mounta
2	p47 0.596 Mount Kilimanjaro in Tanzania is the highest mountain in Africa, risin	p47 0.862 Mount Kilimanjaro in Tanzania is the highest mountain in Africa, risin
3	p20 0.293 The Sahara is the largest hot desert in the world, covering much of No	p20 0.801 The Sahara is the largest hot desert in the world, covering much of No
4	p22 0.282 The Pacific Ocean is the largest and deepest of Earth's oceans, stretc	p18 0.795 The Amazon rainforest in South America is the largest tropical rainfor
5	p48 0.271 Debate continues over the world's longest river: the Nile in Africa is	p22 0.793 The Pacific Ocean is the largest and deepest of Earth's oceans, stretc

exact_paraphrase

Why was the Great Wall of China built?

all-MiniLM-L6-v2 gold rank: 1 · multilingual-e5-small gold rank: 1

#	all-MiniLM-L6-v2	multilingual-e5-small
1	p15 0.744 The Great Wall of China was built over many centuries to defend Chines	p15 0.908 The Great Wall of China was built over many centuries to defend Chines
2	p14 0.190 The Roman Empire was one of the largest empires in ancient history, ru	p21 0.794 The Nile is a major river in northeastern Africa that flows northward
3	p32 0.150 The World Wide Web is a system of interlinked documents and resources	p14 0.783 The Roman Empire was one of the largest empires in ancient history, ru
4	p42 0.145 The Great Barrier Reef off the coast of Australia is the world's large	p42 0.776 The Great Barrier Reef off the coast of Australia is the world's large
5	p27 0.132 The bank of a river is the sloping land alongside the water channel. R	p08 0.773 Global warming is largely driven by greenhouse gases such as carbon di

semantic What organ pushes blood around the body?

all-MiniLM-L6-v2 gold rank: 1 · multilingual-e5-small gold rank: 1

#	all-MiniLM-L6-v2	multilingual-e5-small
1	p03 0.606 The human heart is a muscular organ that pumps blood through the circu	p03 0.867 The human heart is a muscular organ that pumps blood through the circu
2	p07 0.234 Vaccines train the immune system to recognise and fight specific patho	p07 0.809 Vaccines train the immune system to recognise and fight specific patho
3	p25 0.186 Java is a widely used object-oriented programming language designed to	p04 0.803 DNA, or deoxyribonucleic acid, carries the genetic instructions used i
4	p05 0.170 Gravity is the force of attraction between masses. Isaac Newton descri	p01 0.772 Photosynthesis is the process by which green plants and some other org
5	p21 0.130 The Nile is a major river in northeastern Africa that flows northward	p37 0.767 Electric cars are powered by rechargeable batteries instead of petrol

semantic Why is the planet heating up?

all-MiniLM-L6-v2 gold rank: 1 · multilingual-e5-small gold rank: 1

#	all-MiniLM-L6-v2	multilingual-e5-small
1	p08 0.494 Global warming is largely driven by greenhouse gases such as carbon di	p08 0.883 Global warming is largely driven by greenhouse gases such as carbon di
2	p10 0.372 Mars is the fourth planet from the Sun and is often called the Red Pla	p10 0.865 Mars is the fourth planet from the Sun and is often called the Red Pla
3	p20 0.306 The Sahara is the largest hot desert in the world, covering much of No	p02 0.854 The water cycle describes how water evaporates from oceans and lakes,
4	p02 0.291 The water cycle describes how water evaporates from oceans and lakes,	p38 0.845 Solar panels convert sunlight directly into electricity using photovol
5	p05 0.257 Gravity is the force of attraction between masses. Isaac Newton descri	p20 0.845 The Sahara is the largest hot desert in the world, covering much of No

semantic

Which coding language was named after a comedy troupe?

all-MiniLM-L6-v2 gold rank: 1 · multilingual-e5-small gold rank: 1

#	all-MiniLM-L6-v2	multilingual-e5-small
1	p23 0.446 Python is a popular high-level programming language known for its read	p23 0.834 Python is a popular high-level programming language known for its read
2	p44 0.213 William Shakespeare was an English playwright and poet regarded as one	p25 0.761 Java is a widely used object-oriented programming language designed to
3	p25 0.163 Java is a widely used object-oriented programming language designed to	p44 0.745 William Shakespeare was an English playwright and poet regarded as one
4	p50 0.099 In 1903 the Wright brothers achieved the first controlled, powered, su	p32 0.743 The World Wide Web is a system of interlinked documents and resources
5	p32 0.094 The World Wide Web is a system of interlinked documents and resources	p46 0.736 The Pythagorean theorem states that in a right-angled triangle the squ

semantic How can we generate electricity from the sun?

all-MiniLM-L6-v2 gold rank: 1 · multilingual-e5-small gold rank: 1

#	all-MiniLM-L6-v2	multilingual-e5-small
1	p38 0.625 Solar panels convert sunlight directly into electricity using photovol	p38 0.896 Solar panels convert sunlight directly into electricity using photovol
2	p01 0.395 Photosynthesis is the process by which green plants and some other org	p37 0.854 Electric cars are powered by rechargeable batteries instead of petrol
3	p37 0.313 Electric cars are powered by rechargeable batteries instead of petrol	p01 0.849 Photosynthesis is the process by which green plants and some other org
4	p40 0.214 Honeybees are insects that pollinate flowering plants as they collect	p02 0.828 The water cycle describes how water evaporates from oceans and lakes,
5	p02 0.192 The water cycle describes how water evaporates from oceans and lakes,	p05 0.826 Gravity is the force of attraction between masses. Isaac Newton descri

semantic Which insects help flowers reproduce?

all-MiniLM-L6-v2 gold rank: 1 · multilingual-e5-small gold rank: 1

#	all-MiniLM-L6-v2	multilingual-e5-small
1	p40 0.508 Honeybees are insects that pollinate flowering plants as they collect	p40 0.869 Honeybees are insects that pollinate flowering plants as they collect
2	p01 0.252 Photosynthesis is the process by which green plants and some other org	p01 0.781 Photosynthesis is the process by which green plants and some other org
3	p05 0.168 Gravity is the force of attraction between masses. Isaac Newton descri	p18 0.764 The Amazon rainforest in South America is the largest tropical rainfor
4	p18 0.155 The Amazon rainforest in South America is the largest tropical rainfor	p24 0.762 The reticulated python is one of the world's longest snakes, a non-ven
5	p26 0.117 Java is a densely populated island of Indonesia and home to the capita	p04 0.762 DNA, or deoxyribonucleic acid, carries the genetic instructions used i

partial_related

What does caffeine do to the body?

all-MiniLM-L6-v2 gold rank: 1 · multilingual-e5-small gold rank: 1

#	all-MiniLM-L6-v2	multilingual-e5-small
1	p49 0.720 Caffeine is a natural stimulant that acts on the central nervous syste	p49 0.889 Caffeine is a natural stimulant that acts on the central nervous syste
2	p33 0.532 Coffee is a popular brewed drink prepared from roasted coffee beans. I	p33 0.864 Coffee is a popular brewed drink prepared from roasted coffee beans. I
3	p34 0.399 Tea is an aromatic beverage made by pouring hot water over cured leave	p34 0.834 Tea is an aromatic beverage made by pouring hot water over cured leave
4	p03 0.213 The human heart is a muscular organ that pumps blood through the circu	p03 0.815 The human heart is a muscular organ that pumps blood through the circu
5	p01 0.167 Photosynthesis is the process by which green plants and some other org	p07 0.805 Vaccines train the immune system to recognise and fight specific patho

partial_related Famous paintings by Italian artists

all-MiniLM-L6-v2 gold rank: 1 · multilingual-e5-small gold rank: 1

#	all-MiniLM-L6-v2	multilingual-e5-small
1	p43 0.584 The Mona Lisa is a famous portrait painted by the Italian Renaissance	p43 0.863 The Mona Lisa is a famous portrait painted by the Italian Renaissance
2	p44 0.149 William Shakespeare was an English playwright and poet regarded as one	p44 0.787 William Shakespeare was an English playwright and poet regarded as one
3	p24 0.108 The reticulated python is one of the world's longest snakes, a non-ven	p14 0.763 The Roman Empire was one of the largest empires in ancient history, ru
4	p15 0.107 The Great Wall of China was built over many centuries to defend Chines	p15 0.754 The Great Wall of China was built over many centuries to defend Chines
5	p14 0.103 The Roman Empire was one of the largest empires in ancient history, ru	p16 0.751 The French Revolution began in 1789 and overthrew the monarchy in Fran

partial_related

Tall mountains in Africa

all-MiniLM-L6-v2 gold rank: 1 · multilingual-e5-small gold rank: 1

#	all-MiniLM-L6-v2	multilingual-e5-small
1	p47 0.693 Mount Kilimanjaro in Tanzania is the highest mountain in Africa, risin	p47 0.887 Mount Kilimanjaro in Tanzania is the highest mountain in Africa, risin
2	p19 0.507 Mount Everest, on the border of Nepal and Tibet, is the highest mounta	p19 0.842 Mount Everest, on the border of Nepal and Tibet, is the highest mounta
3	p20 0.476 The Sahara is the largest hot desert in the world, covering much of No	p20 0.840 The Sahara is the largest hot desert in the world, covering much of No
4	p48 0.422 Debate continues over the world's longest river: the Nile in Africa is	p48 0.835 Debate continues over the world's longest river: the Nile in Africa is
5	p21 0.326 The Nile is a major river in northeastern Africa that flows northward	p21 0.814 The Nile is a major river in northeastern Africa that flows northward

partial_related

What is the longest river in the world?

all-MiniLM-L6-v2 gold rank: 1 · multilingual-e5-small gold rank: 1

#	all-MiniLM-L6-v2	multilingual-e5-small
1	p48 0.679 Debate continues over the world's longest river: the Nile in Africa is	p48 0.888 Debate continues over the world's longest river: the Nile in Africa is
2	p27 0.425 The bank of a river is the sloping land alongside the water channel. R	p21 0.816 The Nile is a major river in northeastern Africa that flows northward
3	p21 0.413 The Nile is a major river in northeastern Africa that flows northward	p42 0.814 The Great Barrier Reef off the coast of Australia is the world's large
4	p24 0.327 The reticulated python is one of the world's longest snakes, a non-ven	p22 0.812 The Pacific Ocean is the largest and deepest of Earth's oceans, stretc
5	p22 0.292 The Pacific Ocean is the largest and deepest of Earth's oceans, stretc	p47 0.800 Mount Kilimanjaro in Tanzania is the highest mountain in Africa, risin

edge_ambiguous Tell me about oceans and coral reefs

all-MiniLM-L6-v2 gold rank: 1 · multilingual-e5-small gold rank: 1

#	all-MiniLM-L6-v2	multilingual-e5-small
1	p42 0.572 The Great Barrier Reef off the coast of Australia is the world's large	p42 0.870 The Great Barrier Reef off the coast of Australia is the world's large
2	p22 0.425 The Pacific Ocean is the largest and deepest of Earth's oceans, stretc	p22 0.866 The Pacific Ocean is the largest and deepest of Earth's oceans, stretc
3	p41 0.285 Sharks are a group of cartilaginous fish with a powerful sense of smel	p02 0.854 The water cycle describes how water evaporates from oceans and lakes,
4	p08 0.271 Global warming is largely driven by greenhouse gases such as carbon di	p08 0.844 Global warming is largely driven by greenhouse gases such as carbon di
5	p18 0.265 The Amazon rainforest in South America is the largest tropical rainfor	p41 0.831 Sharks are a group of cartilaginous fish with a powerful sense of smel

edge_ambiguous

Popular hot caffeinated beverages

disagreement

all-MiniLM-L6-v2 gold rank: 1 · multilingual-e5-small gold rank: 1

#	all-MiniLM-L6-v2	multilingual-e5-small
1	p34 0.584 Tea is an aromatic beverage made by pouring hot water over cured leave	p33 0.868 Coffee is a popular brewed drink prepared from roasted coffee beans. I
2	p33 0.551 Coffee is a popular brewed drink prepared from roasted coffee beans. I	p34 0.830 Tea is an aromatic beverage made by pouring hot water over cured leave
3	p49 0.462 Caffeine is a natural stimulant that acts on the central nervous syste	p49 0.827 Caffeine is a natural stimulant that acts on the central nervous syste
4	p47 0.156 Mount Kilimanjaro in Tanzania is the highest mountain in Africa, risin	p20 0.766 The Sahara is the largest hot desert in the world, covering much of No
5	p20 0.127 The Sahara is the largest hot desert in the world, covering much of No	p50 0.762 In 1903 the Wright brothers achieved the first controlled, powered, su

edge_ambiguous Facts about famous mountains

all-MiniLM-L6-v2 gold rank: 1 · multilingual-e5-small gold rank: 1

#	all-MiniLM-L6-v2	multilingual-e5-small
1	p47 0.426 Mount Kilimanjaro in Tanzania is the highest mountain in Africa, risin	p47 0.847 Mount Kilimanjaro in Tanzania is the highest mountain in Africa, risin
2	p19 0.423 Mount Everest, on the border of Nepal and Tibet, is the highest mounta	p19 0.845 Mount Everest, on the border of Nepal and Tibet, is the highest mounta
3	p20 0.294 The Sahara is the largest hot desert in the world, covering much of No	p15 0.810 The Great Wall of China was built over many centuries to defend Chines
4	p15 0.222 The Great Wall of China was built over many centuries to defend Chines	p20 0.809 The Sahara is the largest hot desert in the world, covering much of No
5	p48 0.218 Debate continues over the world's longest river: the Nile in Africa is	p43 0.803 The Mona Lisa is a famous portrait painted by the Italian Renaissance

edge_ambiguous

Widely used programming languages

disagreement

all-MiniLM-L6-v2 gold rank: 1 · multilingual-e5-small gold rank: 1

#	all-MiniLM-L6-v2	multilingual-e5-small
1	p23 0.503 Python is a popular high-level programming language known for its read	p25 0.870 Java is a widely used object-oriented programming language designed to
2	p25 0.468 Java is a widely used object-oriented programming language designed to	p23 0.866 Python is a popular high-level programming language known for its read
3	p32 0.303 The World Wide Web is a system of interlinked documents and resources	p31 0.801 Machine learning is a branch of artificial intelligence in which compu
4	p14 0.174 The Roman Empire was one of the largest empires in ancient history, ru	p32 0.794 The World Wide Web is a system of interlinked documents and resources
5	p29 0.169 Cloud computing delivers computing services such as storage and proces	p44 0.789 William Shakespeare was an English playwright and poet regarded as one

edge_acronym What is NASA?

all-MiniLM-L6-v2 gold rank: 1 · multilingual-e5-small gold rank: 1

#	all-MiniLM-L6-v2	multilingual-e5-small
1	p12 0.775 NASA, the National Aeronautics and Space Administration, is the United	p12 0.896 NASA, the National Aeronautics and Space Administration, is the United
2	p10 0.362 Mars is the fourth planet from the Sun and is often called the Red Pla	p10 0.810 Mars is the fourth planet from the Sun and is often called the Red Pla
3	p09 0.261 On 20 July 1969, the Apollo 11 mission landed the first humans on the	p30 0.805 A cloud is a visible mass of water droplets or ice crystals suspended
4	p05 0.229 Gravity is the force of attraction between masses. Isaac Newton descri	p32 0.804 The World Wide Web is a system of interlinked documents and resources
5	p40 0.228 Honeybees are insects that pollinate flowering plants as they collect	p05 0.802 Gravity is the force of attraction between masses. Isaac Newton descri

edge_acronym

The American government space agency

all-MiniLM-L6-v2 gold rank: 1 · multilingual-e5-small gold rank: 1

#	all-MiniLM-L6-v2	multilingual-e5-small
1	p12 0.749 NASA, the National Aeronautics and Space Administration, is the United	p12 0.896 NASA, the National Aeronautics and Space Administration, is the United
2	p09 0.203 On 20 July 1969, the Apollo 11 mission landed the first humans on the	p05 0.797 Gravity is the force of attraction between masses. Isaac Newton descri
3	p10 0.180 Mars is the fourth planet from the Sun and is often called the Red Pla	p50 0.788 In 1903 the Wright brothers achieved the first controlled, powered, su
4	p32 0.178 The World Wide Web is a system of interlinked documents and resources	p11 0.778 A black hole is a region of spacetime where gravity is so strong that
5	p29 0.172 Cloud computing delivers computing services such as storage and proces	p10 0.775 Mars is the fourth planet from the Sun and is often called the Red Pla

edge_acronym

What does DNA do in living things?

all-MiniLM-L6-v2 gold rank: 1 · multilingual-e5-small gold rank: 1

#	all-MiniLM-L6-v2	multilingual-e5-small
1	p04 0.673 DNA, or deoxyribonucleic acid, carries the genetic instructions used i	p04 0.884 DNA, or deoxyribonucleic acid, carries the genetic instructions used i
2	p03 0.289 The human heart is a muscular organ that pumps blood through the circu	p01 0.822 Photosynthesis is the process by which green plants and some other org
3	p01 0.285 Photosynthesis is the process by which green plants and some other org	p07 0.807 Vaccines train the immune system to recognise and fight specific patho
4	p07 0.203 Vaccines train the immune system to recognise and fight specific patho	p40 0.801 Honeybees are insects that pollinate flowering plants as they collect
5	p41 0.192 Sharks are a group of cartilaginous fish with a powerful sense of smel	p38 0.799 Solar panels convert sunlight directly into electricity using photovol

edge_numeric

What major event happened in 1969?

all-MiniLM-L6-v2 gold rank: 1 · multilingual-e5-small gold rank: 1

#	all-MiniLM-L6-v2	multilingual-e5-small
1	p09 0.290 On 20 July 1969, the Apollo 11 mission landed the first humans on the	p09 0.788 On 20 July 1969, the Apollo 11 mission landed the first humans on the
2	p50 0.260 In 1903 the Wright brothers achieved the first controlled, powered, su	p36 0.755 The modern Olympic Games are an international sporting event held ever
3	p16 0.250 The French Revolution began in 1789 and overthrew the monarchy in Fran	p50 0.750 In 1903 the Wright brothers achieved the first controlled, powered, su
4	p13 0.198 World War II ended in 1945 after six years of global conflict. Victory	p17 0.745 The Industrial Revolution was a period of rapid technological and econ
5	p36 0.191 The modern Olympic Games are an international sporting event held ever	p21 0.735 The Nile is a major river in northeastern Africa that flows northward

edge_numeric

What is the value of pi to five decimal places?

all-MiniLM-L6-v2 gold rank: 1 · multilingual-e5-small gold rank: 1

#	all-MiniLM-L6-v2	multilingual-e5-small
1	p45 0.700 Pi is the ratio of a circle's circumference to its diameter, approxima	p45 0.856 Pi is the ratio of a circle's circumference to its diameter, approxima
2	p46 0.100 The Pythagorean theorem states that in a right-angled triangle the squ	p23 0.800 Python is a popular high-level programming language known for its read
3	p22 0.094 The Pacific Ocean is the largest and deepest of Earth's oceans, stretc	p46 0.779 The Pythagorean theorem states that in a right-angled triangle the squ
4	p43 0.086 The Mona Lisa is a famous portrait painted by the Italian Renaissance	p02 0.773 The water cycle describes how water evaporates from oceans and lakes,
5	p23 0.082 Python is a popular high-level programming language known for its read	p06 0.769 According to Albert Einstein's theory of relativity, the speed of ligh

edge_numeric

Which world war ended in 1945?

all-MiniLM-L6-v2 gold rank: 1 · multilingual-e5-small gold rank: 1

#	all-MiniLM-L6-v2	multilingual-e5-small
1	p13 0.759 World War II ended in 1945 after six years of global conflict. Victory	p13 0.902 World War II ended in 1945 after six years of global conflict. Victory
2	p32 0.183 The World Wide Web is a system of interlinked documents and resources	p32 0.776 The World Wide Web is a system of interlinked documents and resources
3	p22 0.150 The Pacific Ocean is the largest and deepest of Earth's oceans, stretc	p15 0.762 The Great Wall of China was built over many centuries to defend Chines
4	p36 0.117 The modern Olympic Games are an international sporting event held ever	p36 0.757 The modern Olympic Games are an international sporting event held ever
5	p02 0.116 The water cycle describes how water evaporates from oceans and lakes,	p16 0.743 The French Revolution began in 1789 and overthrew the monarchy in Fran

edge_numeric

In what year was the first powered airplane flight?

all-MiniLM-L6-v2 gold rank: 1 · multilingual-e5-small gold rank: 1

#	all-MiniLM-L6-v2	multilingual-e5-small
1	p50 0.680 In 1903 the Wright brothers achieved the first controlled, powered, su	p50 0.862 In 1903 the Wright brothers achieved the first controlled, powered, su
2	p09 0.350 On 20 July 1969, the Apollo 11 mission landed the first humans on the	p09 0.765 On 20 July 1969, the Apollo 11 mission landed the first humans on the
3	p12 0.172 NASA, the National Aeronautics and Space Administration, is the United	p17 0.755 The Industrial Revolution was a period of rapid technological and econ
4	p17 0.149 The Industrial Revolution was a period of rapid technological and econ	p37 0.746 Electric cars are powered by rechargeable batteries instead of petrol
5	p13 0.133 World War II ended in 1945 after six years of global conflict. Victory	p32 0.737 The World Wide Web is a system of interlinked documents and resources

edge_type how does fotosynthesis wrok in plants

all-MiniLM-L6-v2 gold rank: 1 · multilingual-e5-small gold rank: 1

#	all-MiniLM-L6-v2	multilingual-e5-small
1	p01 0.583 Photosynthesis is the process by which green plants and some other org	p01 0.869 Photosynthesis is the process by which green plants and some other org
2	p38 0.264 Solar panels convert sunlight directly into electricity using photovol	p02 0.809 The water cycle describes how water evaporates from oceans and lakes,
3	p40 0.194 Honeybees are insects that pollinate flowering plants as they collect	p38 0.807 Solar panels convert sunlight directly into electricity using photovol
4	p04 0.167 DNA, or deoxyribonucleic acid, carries the genetic instructions used i	p46 0.792 The Pythagorean theorem states that in a right-angled triangle the squ
5	p39 0.141 Recycling is the process of collecting and reprocessing used materials	p27 0.788 The bank of a river is the sloping land alongside the water channel. R

edge_typo the tallest montain in the wrld disagreement

all-MiniLM-L6-v2 gold rank: 1 · multilingual-e5-small gold rank: 2

#	all-MiniLM-L6-v2	multilingual-e5-small
1	p19 0.426 Mount Everest, on the border of Nepal and Tibet, is the highest mounta	p47 0.807 Mount Kilimanjaro in Tanzania is the highest mountain in Africa, risin
2	p47 0.422 Mount Kilimanjaro in Tanzania is the highest mountain in Africa, risin	p19 0.799 Mount Everest, on the border of Nepal and Tibet, is the highest mounta
3	p20 0.231 The Sahara is the largest hot desert in the world, covering much of No	p50 0.754 In 1903 the Wright brothers achieved the first controlled, powered, su
4	p14 0.208 The Roman Empire was one of the largest empires in ancient history, ru	p48 0.752 Debate continues over the world's longest river: the Nile in Africa is
5	p22 0.186 The Pacific Ocean is the largest and deepest of Earth's oceans, stretc	p09 0.748 On 20 July 1969, the Apollo 11 mission landed the first humans on the

edge_typo explain the pythagorean theorem

all-MiniLM-L6-v2 gold rank: 1 · multilingual-e5-small gold rank: 1

#	all-MiniLM-L6-v2	multilingual-e5-small
1	p46 0.717 The Pythagorean theorem states that in a right-angled triangle the squ	p46 0.894 The Pythagorean theorem states that in a right-angled triangle the squ
2	p45 0.293 Pi is the ratio of a circle's circumference to its diameter, approxima	p03 0.816 The human heart is a muscular organ that pumps blood through the circu
3	p05 0.181 Gravity is the force of attraction between masses. Isaac Newton descri	p45 0.796 Pi is the ratio of a circle's circumference to its diameter, approxima
4	p03 0.157 The human heart is a muscular organ that pumps blood through the circu	p07 0.794 Vaccines train the immune system to recognise and fight specific patho
5	p06 0.134 According to Albert Einstein's theory of relativity, the speed of ligh	p24 0.793 The reticulated python is one of the world's longest snakes, a non-ven

edge_lexical_trap

Using the Python programming language for data science

all-MiniLM-L6-v2 gold rank: 1 · multilingual-e5-small gold rank: 1

#	all-MiniLM-L6-v2	multilingual-e5-small
1	p23 0.628 Python is a popular high-level programming language known for its read	p23 0.875 Python is a popular high-level programming language known for its read
2	p25 0.227 Java is a widely used object-oriented programming language designed to	p31 0.818 Machine learning is a branch of artificial intelligence in which compu
3	p12 0.188 NASA, the National Aeronautics and Space Administration, is the United	p25 0.811 Java is a widely used object-oriented programming language designed to
4	p31 0.185 Machine learning is a branch of artificial intelligence in which compu	p45 0.801 Pi is the ratio of a circle's circumference to its diameter, approxima
5	p24 0.181 The reticulated python is one of the world's longest snakes, a non-ven	p02 0.795 The water cycle describes how water evaporates from oceans and lakes,

edge_lexical_trap The grassy bank at the edge of a river

all-MiniLM-L6-v2 gold rank: 1 · multilingual-e5-small gold rank: 1

#	all-MiniLM-L6-v2	multilingual-e5-small
1	p27 0.720 The bank of a river is the sloping land alongside the water channel. R	p27 0.873 The bank of a river is the sloping land alongside the water channel. R
2	p21 0.412 The Nile is a major river in northeastern Africa that flows northward	p21 0.819 The Nile is a major river in northeastern Africa that flows northward
3	p28 0.404 A bank is a financial institution that accepts deposits from customers	p28 0.808 A bank is a financial institution that accepts deposits from customers
4	p48 0.357 Debate continues over the world's longest river: the Nile in Africa is	p48 0.778 Debate continues over the world's longest river: the Nile in Africa is
5	p46 0.203 The Pythagorean theorem states that in a right-angled triangle the squ	p02 0.773 The water cycle describes how water evaporates from oceans and lakes,

edge_lexical_trap

Where can I deposit money and open an account?

all-MiniLM-L6-v2 gold rank: 1 · multilingual-e5-small gold rank: 1

#	all-MiniLM-L6-v2	multilingual-e5-small
1	p28 0.543 A bank is a financial institution that accepts deposits from customers	p28 0.864 A bank is a financial institution that accepts deposits from customers
2	p27 0.210 The bank of a river is the sloping land alongside the water channel. R	p27 0.813 The bank of a river is the sloping land alongside the water channel. R
3	p11 0.207 A black hole is a region of spacetime where gravity is so strong that	p21 0.787 The Nile is a major river in northeastern Africa that flows northward
4	p05 0.079 Gravity is the force of attraction between masses. Isaac Newton descri	p03 0.785 The human heart is a muscular organ that pumps blood through the circu
5	p43 0.075 The Mona Lisa is a famous portrait painted by the Italian Renaissance	p29 0.782 Cloud computing delivers computing services such as storage and proces

edge_lexical_trap The island of Java in Indonesia

all-MiniLM-L6-v2 gold rank: 1 · multilingual-e5-small gold rank: 1

#	all-MiniLM-L6-v2	multilingual-e5-small
1	p26 0.828 Java is a densely populated island of Indonesia and home to the capita	p26 0.907 Java is a densely populated island of Indonesia and home to the capita
2	p25 0.486 Java is a widely used object-oriented programming language designed to	p25 0.821 Java is a widely used object-oriented programming language designed to
3	p22 0.204 The Pacific Ocean is the largest and deepest of Earth's oceans, stretc	p22 0.795 The Pacific Ocean is the largest and deepest of Earth's oceans, stretc
4	p47 0.183 Mount Kilimanjaro in Tanzania is the highest mountain in Africa, risin	p42 0.780 The Great Barrier Reef off the coast of Australia is the world's large
5	p13 0.172 World War II ended in 1945 after six years of global conflict. Victory	p13 0.780 World War II ended in 1945 after six years of global conflict. Victory

edge_lexical_trap

Dark clouds gathering in the sky before rain

all-MiniLM-L6-v2 gold rank: 1 · multilingual-e5-small gold rank: 1

#	all-MiniLM-L6-v2	multilingual-e5-small
1	p30 0.639 A cloud is a visible mass of water droplets or ice crystals suspended	p30 0.857 A cloud is a visible mass of water droplets or ice crystals suspended
2	p02 0.343 The water cycle describes how water evaporates from oceans and lakes,	p02 0.796 The water cycle describes how water evaporates from oceans and lakes,
3	p08 0.183 Global warming is largely driven by greenhouse gases such as carbon di	p11 0.787 A black hole is a region of spacetime where gravity is so strong that
4	p29 0.179 Cloud computing delivers computing services such as storage and proces	p20 0.778 The Sahara is the largest hot desert in the world, covering much of No
5	p11 0.176 A black hole is a region of spacetime where gravity is so strong that	p05 0.770 Gravity is the force of attraction between masses. Isaac Newton descri

edge_negation

A programming language that is not Python

all-MiniLM-L6-v2 gold rank: 2 · multilingual-e5-small gold rank: 2

#	all-MiniLM-L6-v2	multilingual-e5-small
1	p23 Python is a popular high-level programming language known for its read 0.726	p23 Python is a popular high-level programming language known for its read 0.869
2	p25 Java is a widely used object-oriented programming language designed to 0.355	p25 Java is a widely used object-oriented programming language designed to 0.823
3	p24 The reticulated python is one of the world's longest snakes, a non-ven 0.296	p31 Machine learning is a branch of artificial intelligence in which compu 0.810
4	p32 The World Wide Web is a system of interlinked documents and resources 0.162	p45 Pi is the ratio of a circle's circumference to its diameter, approxima 0.797
5	p34 Tea is an aromatic beverage made by pouring hot water over cured leave 0.153	p46 The Pythagorean theorem states that in a right-angled triangle the squ 0.789

edge_negation

Common animals that are not mammals

all-MiniLM-L6-v2 gold rank: 1 · multilingual-e5-small gold rank: 1

#	all-MiniLM-L6-v2	multilingual-e5-small
1	p41 0.327 Sharks are a group of cartilaginous fish with a powerful sense of smel	p41 0.841 Sharks are a group of cartilaginous fish with a powerful sense of smel
2	p24 0.227 The reticulated python is one of the world's longest snakes, a non-ven	p24 0.812 The reticulated python is one of the world's longest snakes, a non-ven
3	p03 0.142 The human heart is a muscular organ that pumps blood through the circu	p40 0.808 Honeybees are insects that pollinate flowering plants as they collect
4	p18 0.134 The Amazon rainforest in South America is the largest tropical rainfor	p03 0.807 The human heart is a muscular organ that pumps blood through the circu
5	p09 0.129 On 20 July 1969, the Apollo 11 mission landed the first humans on the	p37 0.804 Electric cars are powered by rechargeable batteries instead of petrol

edge_negation

Hot drinks that contain no caffeine

disagreement

no relevant passage (unanswerable)

#	all-MiniLM-L6-v2	multilingual-e5-small
1	p34 0.450 Tea is an aromatic beverage made by pouring hot water over cured leave	p33 0.855 Coffee is a popular brewed drink prepared from roasted coffee beans. I
2	p49 0.436 Caffeine is a natural stimulant that acts on the central nervous syste	p49 0.848 Caffeine is a natural stimulant that acts on the central nervous syste
3	p33 0.428 Coffee is a popular brewed drink prepared from roasted coffee beans. I	p34 0.841 Tea is an aromatic beverage made by pouring hot water over cured leave
4	p20 0.118 The Sahara is the largest hot desert in the world, covering much of No	p37 0.776 Electric cars are powered by rechargeable batteries instead of petrol
5	p24 0.111 The reticulated python is one of the world's longest snakes, a non-ven	p20 0.773 The Sahara is the largest hot desert in the world, covering much of No

no_match

Best recipe for chocolate chip cookies

disagreement

no relevant passage (unanswerable)

#	all-MiniLM-L6-v2	multilingual-e5-small
1	p50 0.133 In 1903 the Wright brothers achieved the first controlled, powered, su	p33 0.767 Coffee is a popular brewed drink prepared from roasted coffee beans. I
2	p09 0.114 On 20 July 1969, the Apollo 11 mission landed the first humans on the	p01 0.766 Photosynthesis is the process by which green plants and some other org
3	p40 0.089 Honeybees are insects that pollinate flowering plants as they collect	p34 0.766 Tea is an aromatic beverage made by pouring hot water over cured leave
4	p35 0.062 Chess is a two-player strategy board game played on an eight-by-eight	p23 0.761 Python is a popular high-level programming language known for its read
5	p33 0.056 Coffee is a popular brewed drink prepared from roasted coffee beans. I	p46 0.758 The Pythagorean theorem states that in a right-angled triangle the squ

no_match

How do I change a flat car tire? disagreement

no relevant passage (unanswerable)

#	all-MiniLM-L6-v2	multilingual-e5-small
1	p05 0.108 Gravity is the force of attraction between masses. Isaac Newton descri	p37 0.785 Electric cars are powered by rechargeable batteries instead of petrol
2	p16 0.099 The French Revolution began in 1789 and overthrew the monarchy in Fran	p17 0.775 The Industrial Revolution was a period of rapid technological and econ
3	p04 0.068 DNA, or deoxyribonucleic acid, carries the genetic instructions used i	p27 0.774 The bank of a river is the sloping land alongside the water channel. R
4	p08 0.060 Global warming is largely driven by greenhouse gases such as carbon di	p02 0.771 The water cycle describes how water evaporates from oceans and lakes,
5	p10 0.056 Mars is the fourth planet from the Sun and is often called the Red Pla	p03 0.769 The human heart is a muscular organ that pumps blood through the circu

no_match What is the current stock price of Apple?

no relevant passage (unanswerable)

#	all-MiniLM-L6-v2	multilingual-e5-small
1	p25 0.132 Java is a widely used object-oriented programming language designed to	p25 0.741 Java is a widely used object-oriented programming language designed to
2	p29 0.089 Cloud computing delivers computing services such as storage and proces	p38 0.738 Solar panels convert sunlight directly into electricity using photovol
3	p47 0.085 Mount Kilimanjaro in Tanzania is the highest mountain in Africa, risin	p29 0.737 Cloud computing delivers computing services such as storage and proces
4	p06 0.084 According to Albert Einstein's theory of relativity, the speed of ligh	p17 0.736 The Industrial Revolution was a period of rapid technological and econ
5	p19 0.061 Mount Everest, on the border of Nepal and Tibet, is the highest mounta	p37 0.735 Electric cars are powered by rechargeable batteries instead of petrol

no_match Explain the rules of cricket batting

no relevant passage (unanswerable)

#	all-MiniLM-L6-v2	multilingual-e5-small
1	p35 0.230 Chess is a two-player strategy board game played on an eight-by-eight	p35 0.814 Chess is a two-player strategy board game played on an eight-by-eight
2	p07 0.215 Vaccines train the immune system to recognise and fight specific patho	p03 0.810 The human heart is a muscular organ that pumps blood through the circu
3	p40 0.201 Honeybees are insects that pollinate flowering plants as they collect	p07 0.806 Vaccines train the immune system to recognise and fight specific patho
4	p34 0.120 Tea is an aromatic beverage made by pouring hot water over cured leave	p46 0.791 The Pythagorean theorem states that in a right-angled triangle the squ
5	p05 0.111 Gravity is the force of attraction between masses. Isaac Newton descri	p36 0.790 The modern Olympic Games are an international sporting event held ever

no_match How do I house-train a puppy? **disagreement**

no relevant passage (unanswerable)

#	all-MiniLM-L6-v2	multilingual-e5-small
1	p31 0.135 Machine learning is a branch of artificial intelligence in which compu	p07 0.795 Vaccines train the immune system to recognise and fight specific patho
2	p50 0.109 In 1903 the Wright brothers achieved the first controlled, powered, su	p37 0.781 Electric cars are powered by rechargeable batteries instead of petrol
3	p07 0.104 Vaccines train the immune system to recognise and fight specific patho	p24 0.779 The reticulated python is one of the world's longest snakes, a non-ven
4	p37 0.088 Electric cars are powered by rechargeable batteries instead of petrol	p03 0.773 The human heart is a muscular organ that pumps blood through the circu
5	p03 0.048 The human heart is a muscular organ that pumps blood through the circu	p46 0.772 The Pythagorean theorem states that in a right-angled triangle the squ

8. Caveats

- **Synthetic data.** A hand-authored 50-passage corpus is small; absolute metrics are optimistic and category-level numbers rest on a few queries each. Use the *relative* comparison, not absolute scores.
- **CPU timing.** Latencies are CPU-only and indicative, not production throughput.
- **English-only.** e5's multilingual capability is untested here; if your real workload is multilingual, re-run with a multilingual corpus before deciding.
- **Swap in real data.** Replace `data/dataset.json` with your own passages/queries/qrels and re-run `evaluate.py` + `report.py` unchanged.

Generated by the embedding comparison harness · raw data in `results.json` / `aggregate.csv` / `per_query.csv`